

Comparison with evaluation and development boards

A single-board computer is a printed circuit board constructed using a single microprocessor/microcontroller and all other peripherals like ADC/DAC, input/outputs, memories, RS485 port, Ethernet port, VGA, RS232 port and GSM/GPRS on the same board. It is a face-less computer mainly for industrial applications that can be completely customised to fit the requirement.

"A single-board computer is a printed circuit board with all the components and wiring to make a computer. Even the ports required to communicate to the outside world are included. What's missing is the chassis—the box that houses the SBC," shares Royson Rajan, vice president, Solusoft Technologies.

The main difference between SBCs, evaluation boards and development boards is that the SBC is concluded as a final product. "The end production/manufacturing of new product includes the same SBC as was used during development, while evaluation and development boards support the development and are not used as the final product," explains Rajesh V., managing director, MRR Embedded Systems. "Evaluation board have limited features to evaluate the process that the board is meant for. For example, if the project includes Ethernet, CAN/LIN, battery charger and SD card/USB, the design engineer needs separate evaluation boards for each of these, to make sure that it works the way he wants before integrating all these together. A development board is used for developing the entire project."

Factor Special Interest Group (SFF-SIG) and is an open standard.

The success of VIA's Mini-ITX has increased the use of SBCs in power supplies, slim-line drivers, Flash-to-IDE adaptors and many handy computing components. "Mini-ITX 17x17cm² is targeted at new-generation, low-power, small-size, infotainment systems to enable innovative developments," says Sathya. Some of the latest offerings include SBCs using Intel's Atom CPU. It's a huge leap forward exploding its popularity between home users and hobbyists. Many SBC manufacturers have realised the potential in Mini ITX form factor and offer several products in this form factor using either VIA's chips or Intel's chipsets.

EBX: Fit for Pentium

The Embedded Board eXPandable (EBX) offers suitable features to support embedded designs with high performance. It serves as the right mix of size (20.32x14.6 cm²), economy and functionality, which makes it suitable to embed Pentium CPUs. Many OEMs are able to include high-performance graphics and advanced networking on EBX.

Derived from Ampro Little Board form-factor, EBX combines a stand-

ard footprint with open interfaces. The EBX standard is the result of collaboration between industry leaders, Motorola and Ampro, to unify the embedded computing industry on a small-footprint embedded single-board computer standard. This form-factor is small enough for deeply embedded applications, yet large enough to contain the functions of a full embedded computer system (CPU, memory, mass storage interfaces, display controller, serial/parallel ports and other system functions).

Built-in display function

Today's embedded systems are incomplete without a human-machine interface, where use of high-end displays like VGA, TFT and touch screens is rapidly growing. As these display interfaces become affordable, the usage becomes more lenient. "To support the fully-functional, feature-rich product development, SBC manufacturers have started integrating display as well as link to inputs/control," shares Patel.

"SBCs traditionally had 7-segment LED displays useful for control applications. Easy and low-cost availability of LCDs, TFT displays as well as touchscreen panels have made their way on SBCs. These are suitable for ap-

Advantages of SBCs

1. Single board with features integrated in a compact size
2. Availability of backplanes and several slot configurations such as VME, PCI, ISA and Ethernet
3. Availability of more slots for interconnect
4. Minimal component count
5. Quick time-to-market
6. More power-efficient, lighter and more reliable

plications like automation, automotive electronics, medical devices and GPS navigation systems," explains Patel.

Display panel manufacturers are realising a large and growing market in industrial areas. The display units are made rugged to survive and work in temperature range of -25°C to +85°C. Such rugged systems can be fitted into industrial computers and automation.

Open source SBCs

Open source software and hardware have become increasingly famous for development as these offer all the flexibility to change and use the way a designer wants to. There are several single-board computers that come under open source hardware category, allowing design development with reduced time-to-market and low cost.

One of the commonly used open source SBCs is the Beagle Board (7.6x7.6cm²) based on Texas Instrument's OMAP3530 system-on-chip. The USB-powered Beagle Board offers a low-cost, low-power solution that unleashes laptop-like performance and expandability without the bulk expense or noise of a typical desktop machine.

Another widely used SBC is OMAP L138 dual-core processor based Hawk Board that also includes floating-point DSP C674x application from TI. Hawk Board is an ultimate tool for digital media and storage applications with a wide range of peripheral support. An SBC named Gumstix, of the size of